

Torque ripple minimisation of permanent magnet synchronous motor using a new proportional resonant controller

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Abstract: When applying the permanent magnetic synchronous motors (PMSM) to refrigerant system, the suppression of speed ripples of the PMSM in low-feed range becomes the focus of consideration. The authors start with the generation mechanism and the periodical characteristics of the speed ripples to propose an adaptive control method. In the proposed method, the authors place a frequency variable resonance controller in parallel with the conditional proportion and integral (PI) controller to form a PI-resonance (PI-RES) controller. The compensation torque current generated by the resonance controller and the main reference current generated by the PI controller constitute the reference torque current. Because of the existence of the compensation torque current, the electromagnetic torque can follow the variation of the load torque much better, thus the speed ripples are suppressed. Performance indices of the conventional PI controller and the PI-RES controller are compared and evaluated through experimental investigations. The results validate the effectiveness of the proposed method.

1 Introduction

The permanent magnet synchronous motor (PMSM) has been widely used in many industrial applications because of its high efficiency, fewer parts, light weight and small size [1]. With the variable-speed refrigerant systems performance and energy efficiency is becoming increasingly demanding in recent years, the traditional induction motor used in compressor has gradually been replaced by PMSM. However, there are still some limits for the popularisation and application of this variable speed compressor, one of the most serious is the low-speed range of the speed fluctuations and the resulting low-frequency noise and vibration problems.

To overcome the vibration problem of compressor, the compressor operates in high speed range where the rotor and piston inertia relieve the vibration phenomenon. However, this high speed operation decreases the overall system efficiency since it has to be ON/OFF control to reach the demanded temperature [2]. Otherwise, which is of our interest, concentrates on using an additional control effort to compensate these periodic torque pulsations. The predefined load torque compensator is a popular method used to minimise the speed ripple and vibration at low speed [3, 4]. However, in such a method, sufficiently accurate information of the PMSM parameters, in particular the characteristics of torque ripples are required, and a small error or variations in parameters can lead to an even higher torque ripple due to the open-loop feed-forward control [5]. Various methods have been proposed for instantaneous torque estimation [6–8]. However, since the environment of high temperature and pressure inside the compressor, the sensorless methods are normally has to be used [9–11]. The load torque observer diverges with the senseless method which has originally speed ripple and position error varying on electrical noise and the parameter variation by the saturation and measurement error [12]. Gu *et al.* [12] present an online load torque compensator, but need complex fast Fourier transform (FFT) transform. In [5] iterative learning controller is applied in conjunction with the conventional proportional-integral (PI) speed controller to minimise the speed ripples.

In this paper, a frequency variable resonance controller is applied in conjunction with the conventional proportional-integral (PI) speed

controller as a PI-RES speed controller, which provides the reference of torque current. Compensation torque current generated by the resonance controller, the proposed controller that together with the main reference current is utilised to minimise speed ripples. PI-RES controllers are also used in the inner control loops to generate the control voltages to obtain pulse-width-modulated signals. The comparison performances between the two control method by using conventional PI controller and PI-RES controller have been evaluated through experimental investigations.

The remainder of this paper is organised as follows. First, the mathematical model of PMSM is reviewed in Section 2. The torque characteristics of the compressor and the reasons of generating speed ripple are discussed in Section 3. The proposed control method is explained, and PMSM drive setup used in experiments is introduced in Section 5. Section 6 presents and discusses the experimental results. Some conclusions are drawn in Section 7.

2 Mathematical model of PMSM

With assumptions that the PMSM is unsaturated and eddy currents and hysteresis losses are negligible. In the synchronous $d-q$ frame, the voltage equations can be expressed as.

$$\begin{cases} v_d = R_s i_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q \\ v_q = R_s i_q + L_q \frac{di_q}{dt} + \omega_e L_d i_d + \omega_e \lambda_m \end{cases} \quad (1)$$

where v_d is the d -axis voltage, v_q is the q -axis voltage, R_s is the stator resistance, i_d is the d -axis current, i_q is the q -axis current, L_d is the d -axis inductance, L_q is the q -axis inductance, ω_e is the synchronous electric speed and λ_m is the flux linkage of the permanent magnet mounted on the shaft rotor. Using the method of field-oriented control of the PMSM, the d -axis current is

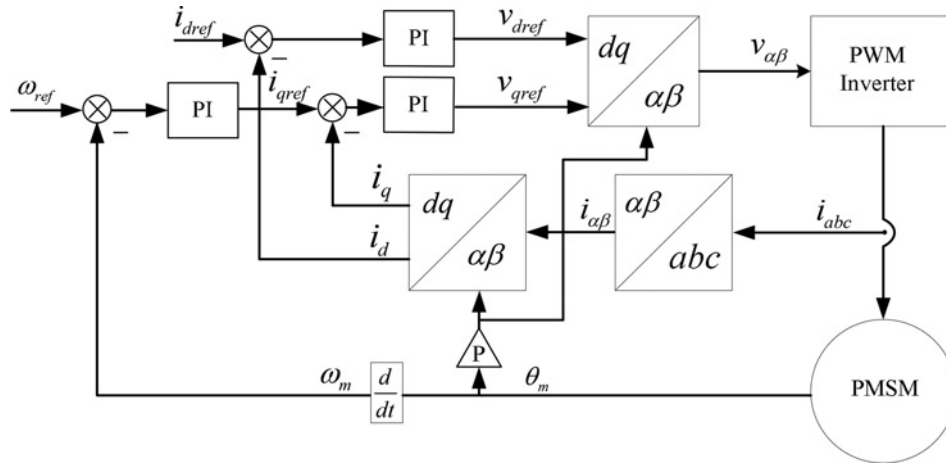


Fig. 1 Classic dual loop field-oriented control of the PMSM by using PI controller

usually controlled to be zero. The electro-magnetic torque is

$$T_e = \frac{3P}{2} \lambda_m i_q = k_t i_q \quad (2)$$

where T_e the electro-magnetic torque of the motor is, P is the number of poles of the motor and k_t is the torque constant. The mechanical speed and position of the motor are expressed as

$$\frac{d\omega_m}{dt} = \frac{1}{J_m} (T_e - T_L - B_m \omega_m) \quad (3)$$

And

$$\frac{d\theta_m}{dt} = \omega_m \quad (4)$$

where, ω_m is the mechanical speed, θ_m is the mechanical position, J_m is the inertia, T_L is the external load and B_m is the viscous coefficient.

In a classic dual loop field-oriented control of the PMSM, the d -axes, and q -axes currents can be controlled separately in the inner loop by using PI controller for maintaining a desired machine flux level and generating desired machine torque that is determined by the outer speed loop PI controller. The complete control diagram is shown in Fig. 1.

3 Speed ripple characteristics of PMSM with compressor load

Normally viscosity coefficient B_m is very small, can be neglected. Using differentiator s instead of (d/dt) , from (3), the plant transfer function between the motor speed and the torque is

$$\omega_m(s) = \frac{\Delta T_m}{J_m s} \quad (5)$$

where

$$\Delta T_m = T_e - T_L \quad (6)$$

It can be seen that the speed would oscillate at the same harmonic frequencies as those of ΔT_m , especially at low operating speeds. It is imperative that to minimise the speed ripples the sources of these speed oscillations – the error torque pulsations ΔT_m need to be minimised.

As is well known, the compressor have position-dependent load torque, the torque varies dramatically for the different position of rotor, and the torque ripple frequency varies for the different rotor speed. In the case of the paper, the compressor's load locus for rotor running at 62.83 rad/s is shown in Fig. 2a. If ignore the dc term, it can be seen the torque rippled at 20 Hz, when the rotor running at 10 Hz, from the FFT analysis in Fig. 2b. Hence, there have two torque ripple period in a mechanical revolving period.

In tradition case, the load torque usually is constant in steady-state condition. The outer speed loop can achieve good performance, either in steady-state or dynamic-state by using PI controller. However, with the position-dependent load torque, the variety of the ripped term of torque is continual and nearly sinusoidal with twice rotor frequency as shown in Fig. 2a.

The tradition outer speed loop by using a PI controller can be shown as (Fig. 3):

Where K_p and K_i , respectively, is the proportional and integral coefficient of speed loop PI controller. T_{di} is the delay of inner loop. ω_{ref} is the speed reference, it is usually constant.

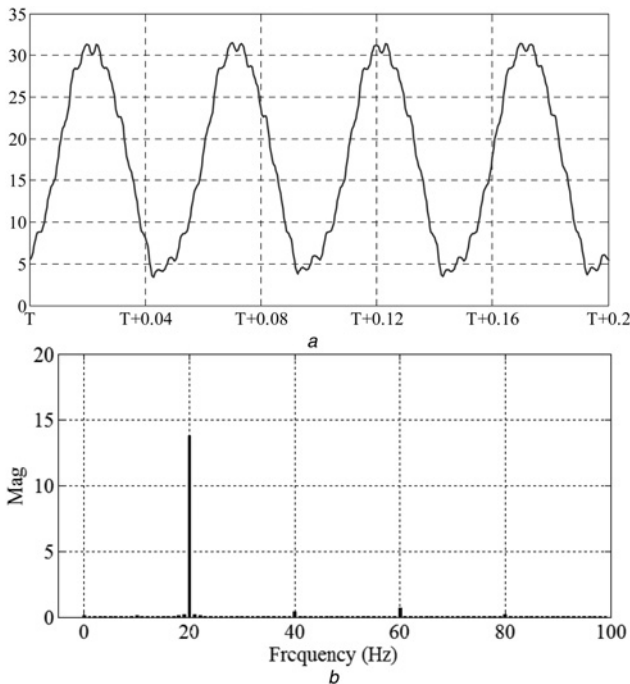


Fig. 2 Measured load locus of the compressor for rotor running at 62.83 rad/s

a Compressor load locus

b FFT analysis of the compressor load locus

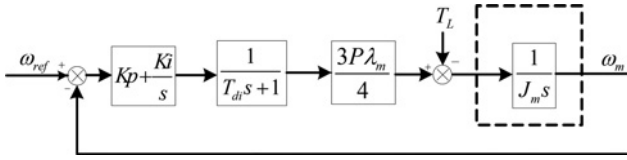


Fig. 3 Control block diagram of the tradition outer speed loop by using a PI controller

Because of the limited bandwidth of the speed loop with PI controller, and standard integrators that can achieve good none-error control just at zero frequency but others frequency. It is hard to achieve $\Delta T_m \simeq 0$. ΔT_m will ripple with twice rotor frequency. The speed will ripple with twice rotor frequency. Moreover, as shown in (5), because of the term of inertia, the amplitude of the ripple speed is inverse proportional to the ripple frequency, and then inverse proportional to the rotor speed. Hence, in the low speed, the speed would severely ripple. There will be inevitable vibration and the noise in low speed of the compressor, which is the almost operation condition because of the efficiency. Hence it is necessary to suppress the ripple of ΔT_m in the low speed of the rotor.

4 Proposed scheme for PMSM

4.1 Review of resonant controllers

A PR controller for tracking an AC signal can be expressed in the s -domain as in [13, 14]

$$G_{PR}(s) = K_P + \frac{2K_{ri}\omega_c s}{s^2 + 2\omega_c s + \omega^2} \quad (7)$$

With ω being the frequency of adjusted signal, ω_c is the damping coefficient, will enlarge the band-wide of central frequency of ω and enlarge the PM of control system. $G_{PR}(s)$ provides the infinite gain in open loop at the resonant frequency ω , which assures perfect tracking for components oscillating at ω when implemented in closed loop, such as $G_{PI}(s)$ implemented in the rotating frame with the frequency of ω [14]. If $G_{PR}(s)$ controllers and $G_{PI}(s)$ are employed in parallel for $G_{PI-RES}(s)$, only a single gain K_P should be adjusted [15].

$$G_{PI}(s) = K_P + \frac{K_I}{s} \quad (8)$$

$$G_{PI-RES}(s) = K_P + \frac{K_I}{s} + \frac{2K_{ri}\omega_c s}{s^2 + 2\omega_c s + \omega^2} \quad (9)$$

4.2 Proposed scheme for PMSM

Obviously, because the dominating perturbation is continual and sinusoidal with twice frequency of the rotor, there will be very good by adding a twice rotor frequency resonant controller combines with the traditional PI regulator to form a new PI-RES controller that have good capability used to control the harmonics

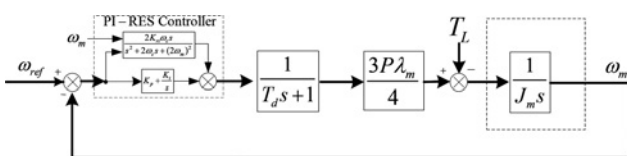


Fig. 4 Control block diagram of outer speed loop by using a PI-RES controller

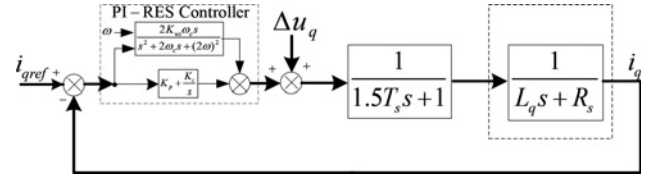


Fig. 5 Control block diagram of inner current loop by using a PI-RES controller

in [14] instead of traditional PI controller as shown in Fig. 4. K_{ri} is the resonance coefficient, and ω_c is the damping coefficient. The resonance term is centred about the second harmonic frequency of rotor which obtained by speed sensorless method or encoder. Compensation torque current generated by the resonance term that together with the main reference current generated by the PI term is utilised to minimise the speed ripples.

With the PI-RES speed controller, the rippled torque current reference which is going to counteract the rippled term of load torque of compressor would be generated. With the same condition, in order to obtain the none-error control of ac term of torque current generated by the resonance compensator, another resonant need added to the inner current loop, as shown in Fig. 5.

The speed of PMSM usually is obtained by (4), the high frequency noise is inevitably generated by the process of differential in a discrete system. In order to suppress the high frequency noise in speed signal, 500 Hz low-pass filter is used. In summary, the complete control diagram of the proposed dual loop field-oriented control of the PMSM with the load of compressor is as shown in Fig. 6.

The control scheme of Fig. 6 consists of two cascaded control loops: an outer speed control loop using a PI-RES regulator to adjust the speed, the PI term is used to obtain good dynamic performance for a speed step, and the PR term is used to eliminate the speed ripple with the frequency of twice ω_m .

An inner current control loop also employing a PI-RES to regulate both dc and ac current in the rotating frame with the frequency of ω_c . The output of the speed controller sets the reference for the torque current.

5 Controller design

5.1 Current control loop design

As discussed above, the current control loop is shown in Fig. 5. It has been demonstrated that the PR controller implemented in stationary reference frame is the same as of PI regulators in rotating frame, these parameters can be tuned with the 'technical optimum' approach [16] using zero to cancel the pole as

$$K_{Pi} = \frac{L_q}{4\xi^2 T_d}, K_{ii} = \frac{K_{Pi} R}{L_q}, K_{Ri} = K_{ii} \quad (10)$$

where, $T_d = 1.5T_s$, $\xi = 0.8$.

Considering the system adopted for the experimental validation (the parameters are given in Table 1), and with the cut-off frequency $\omega_c = 5$ rad/s, the Bode plots of the open loop and closed loop current control are shown in Fig. 7. The bandwidth obtained for difference ω are nearly the same $B_i \simeq 4070$ rad/s and phase margin are 38.4° , respectively.

5.2 Speed control loop design

The speed control loop is shown in Fig. 4. In the speed control loop, the dynamic performance is determined by the proportional term K_{ps} . The integral term and resonance term eliminate the error in steady-state. When $K_{is} = 0$, and $K_{rs} = 0$, the speed loop phase margin is close to 90° . The system will be stable. Hence, if we

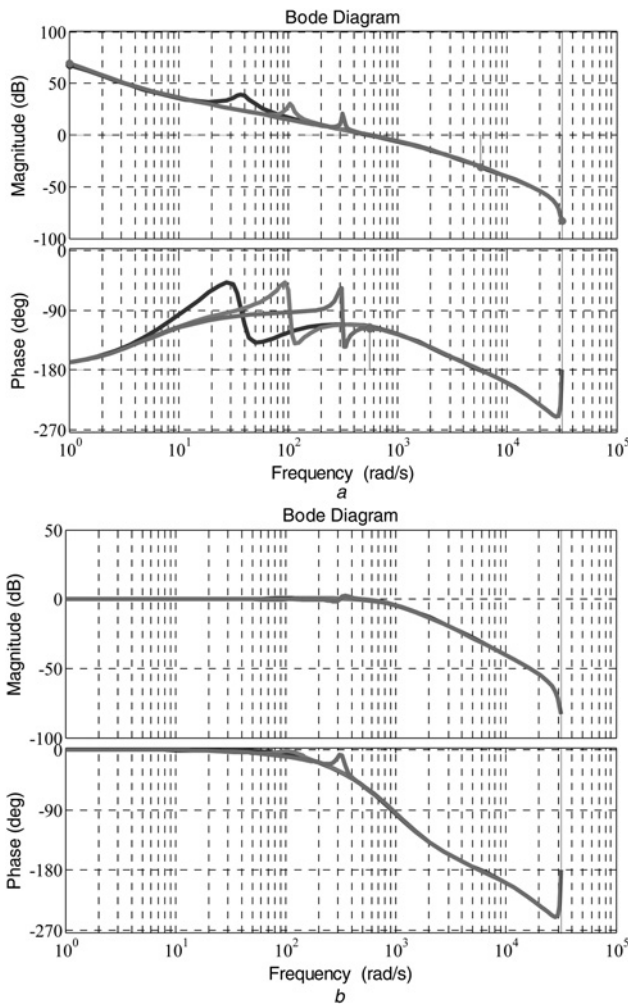


Fig. 8 Bode diagram of current loop for the rotor

a Open-loop Bode diagram of current loop for the rotor running at 180, 500 and 1500rpm
b close-loop Bode diagram of current loop for the rotor running at 180, 500 and 1500rpm



Fig. 9 Photo of the experiment system

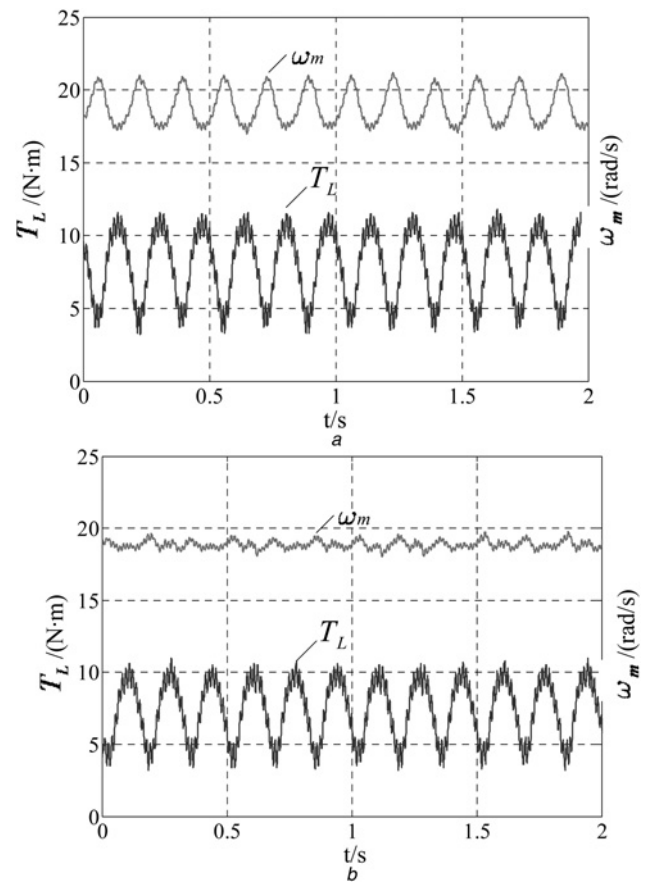


Fig. 10 Speed response when the speed reference of the motor set at 18.85 rad/s (180r/min)

a By using conventional control method
b By using proposed control method

rotor position information. The parameters in experiments are given in Table 1.

The experimental platform configuration is shown in Fig. 9. The main chip of the inverter adopts the TMS320F28335 (TI corporation). The experiments were conducted under different operating conditions, with speed reference at 18.85 rad/s (180 r/min), 52.36 rad/s (500 r/min) and rated speed 157 rad/s (1500 r/min).

Fig. 10 shows the speed response when the speed reference of the motor set at 18.85 rad/s (180 r/min) by using classic dual loop field-oriented control. In this figure, a large speed oscillation can be observed with the peak-peak value of 4.24 rad/s. Figs. 11*a* and *b* present the speed waveforms under the same working condition as Fig. 11 by using the proposed control method. The speed ripple is reduced. And the peak-peak value of the speed is 0.87 rad/s.

Figs. 11*c* and *d* show the same comparison, and the operating reference for these figures are 52.36 rad/s (500 r/min) and 157 rad/s (1500 r/min). It can be seen again that both the proposed schemes are able to restrain the speed ripple with the compress load either in middle speed or in the rated speed.

The dynamic performance of the motor is also compared. Fig. 12, show the speed performance and current performance when the motor accelerate from 0 rad/s to the rated by using conventional method and the proposed method. It can be shown that the introduced resonance part in the speed controller and current controller do not affect the dynamic performance of the motor system with the compress load.

7 Conclusion

This paper has proposed a new control strategy for PMSM with compress load. Frequency variable resonance controller is applied

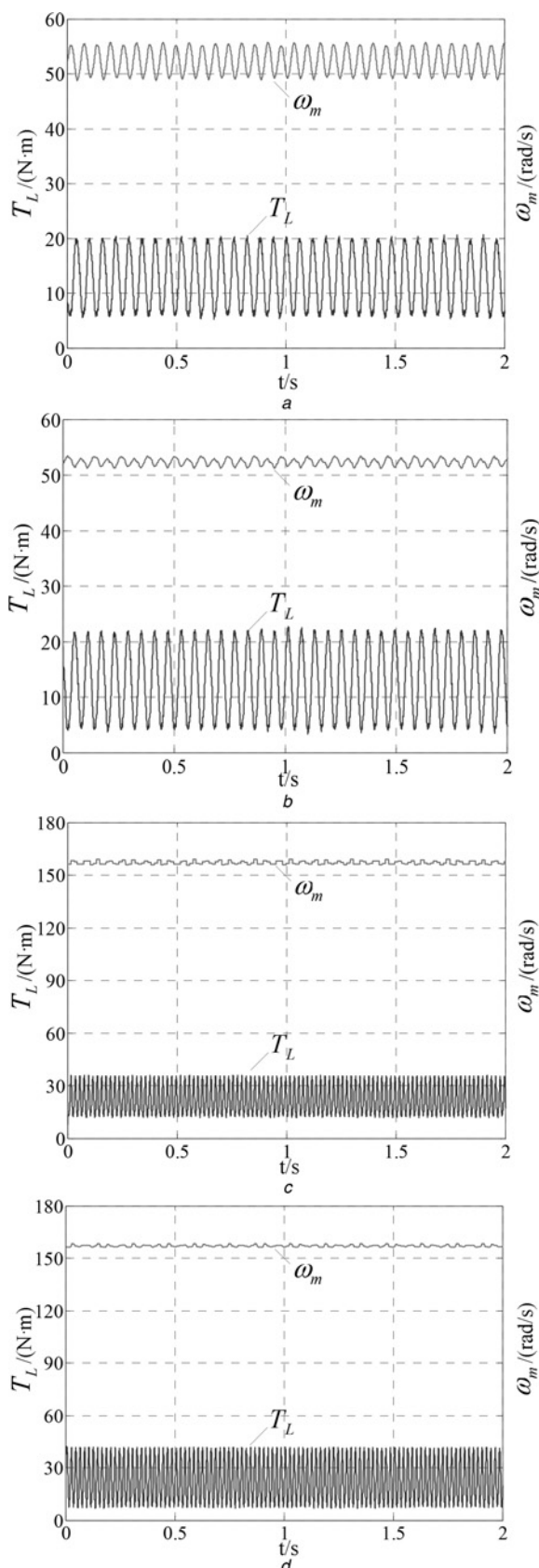


Fig. 11 Speed response motor

a Speed response by using conventional control method (52.36 rad/s (500 r/min))
b Speed response by using proposed control method (52.36 rad/s (500 r/min))
c Speed response by using conventional control method (157 rad/s (1500 r/min))
d Speed response by using proposed control method (157 rad/s (1500 r/min))

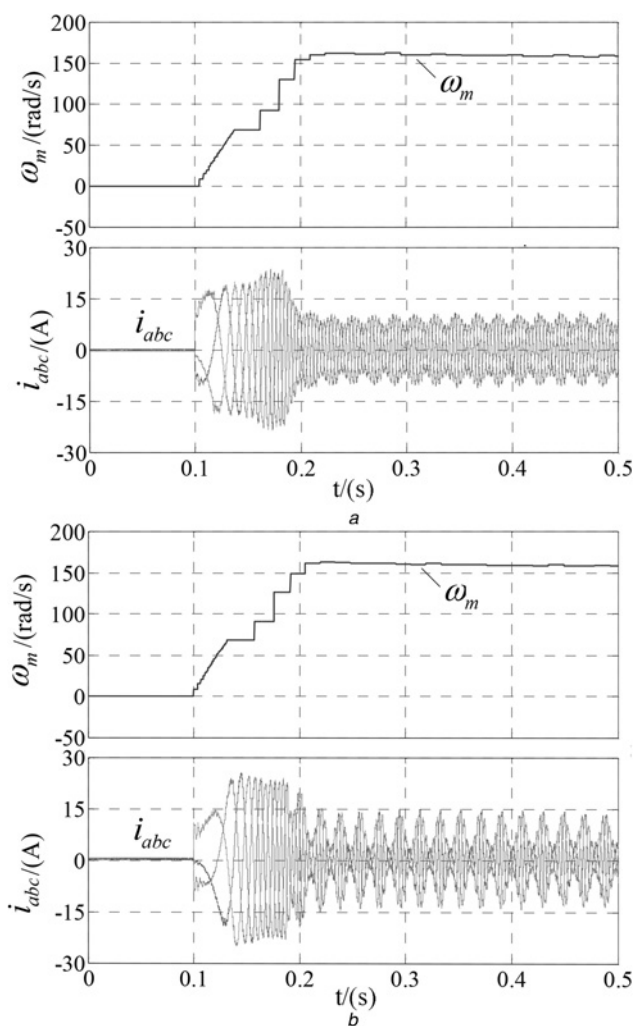


Fig. 12 Dynamic performance of motor

a Dynamic performance by using conventional control method
b Dynamic performance by using proposed control method

in conjunction with the conventional PI as a PI-RES controller to adjust the rippled speed and the current when the load rippled periodically with the speed. The new strategy is verified by experiment. It has been proved the method was effective compared with the traditional method. The ripples have been suppressed in the whole speed range, especially in the low speed.

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